Regularl

Theory of Computation

2-81

Regular Languages

Consider the output sequence for Moore machine for input sequence,

abba Then

90 a	obe	+q1 b	lbe a	obe
0	0	11	1	0

Similarly output sequence for Mealy machine will be

10			91	0bt
0	1	1	1	0

We can note that length of Moore machine is n+1 and that of Mealy machine is n

2.13.3 Method for Conversion of Mealy Machine to Moore Machin

In Moore machine the output is associated with every state and in Mealy machine the output is given along the edge with input symbol. To convert Moore machine to Mealy machine state output symbols are distributed to input symbol paths. But while converting Mealy machine to Moore machine we will create a separate state for every new output symbol and according incoming and outgoing edges are distributed.

Let M =Q,, 8,. qo be a Mealy machine then there exists a Moore machine M equivalent to M.

The Mcan be given as M = QxA, A8[qobl) where bg is an output symbol selected from. We will calculate 8 and as follows-

[qb], a=[8qa), qa] 1[qb]=b 8defines the move made by M on input a.

M on input a.

defines the output made for the corresponding state q Thus we have to calculate 8 and Xfor qo q1q qn on input ag ajdn and should emit the outputs bb b... bn.

Let us understand this method of conversion with the help of some examples

Example 2.80. Convert the following Mealy machine into equivalent Moore machine.

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1/A Fig.2.69